

1. Basic One-Hot Encoding

Use when: Categories are nominal (no order) and data fits in memory.

Purpose: Converts each category into a binary vector (0s and 1s).

2. Pandas `get_dummies()`

Use when: You're working in pandas and want a quick one-hot encode.

Purpose: Simple and fast for small to medium datasets.

3. `OneHotEncoder` from Scikit-learn

Use when: You want to integrate encoding in a ML pipeline or use sparse matrices.

Purpose: More flexible, handles unknown categories, can return sparse matrix.

4. Dummy Encoding (Drop one column)

Use when: Avoiding the dummy variable trap in regression models.

Purpose: Reduces multicollinearity by dropping one encoded column.

5. Binary Encoding (Hybrid of one-hot and hashing)

Use when: High cardinality categories (many unique values).

Purpose: Reduces dimensionality compared to regular one-hot.